



The Scientific Revolution: Galileo, Newton, and the Birth of Modern Science

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Abstract

The scientific revolution was a period between the 15th and 17th centuries, when naturalism in science and modern scientific method arose from the ruins of the world. The different philosophies that were being developed created opportunities for many people to become scientists. Galileo Galilei and Isaac Newton, who have changed the view of science in the eyes of billions of people, beginning the process of science in this world. Examples of great discoveries included by great scientists: Copernicus and Galileo's heliocentric revolution, disproving the traditional belief of Earth as the center of the solar system; Isaac Newton's development of the calculus and laws of motion and gravity; Albert Einstein's theory of relativity, as well as Charles Darwin's study of evolution. This article will cover the life of these scientists, the beginning of science that led to the present era of science, the background that led to the scientific revolution, the beginning of science in this world, as well as the effects that the scientific revolution had on the world, such as science, education, technology as well as life in general.

Key words: Scientific Revolution, Newton, Galileo, Copernicus, Darwin, Relativity, Darwin's theory of evolution, revolution.

Introduction

The "scientific revolution" of the beginning of the nineteenth century is named for the profound changes in the way nature was viewed at the time. A "revolution" is a period of social upheaval and instability when revolutionary events occur, and a new set of institutions begins to emerge. This term, therefore, describes quite accurately what took place in the scientific community following the scientific revolution (James Turner). During the scientific revolution, much of scientific philosophy was attacked in favor of the new methods proposed by Bacon, Galileo, Descartes, and Newton. The importance of experimentation in the scientific method was highlighted. The importance of God was downplayed for the most part, and the power of reason itself (rather than philosophy) gained validity in the sciences (Descartes). The revolution marked numerous intellectual shifts which laid the foundation for modern science. By suggesting that the Earth revolved around the Sun, Nicholas Copernicus' heliocentric model overturned the geocentric viewpoint, influencing future astronomy and science. Johannes Kepler built upon Copernicus' work by postulating elliptical orbits and introducing mathematical analysis and empirical facts, leading to his laws of planetary motion. Galileo Galilei furthered the development. Descartes supported the Copernican revolution by advocating rational deduction like Kepler's ideas, moving away from philosophical conjecture and toward experimental observation (Richard Schwab). Isaac Newton developed the laws of motion and universal gravitation by synthesizing the work of his predecessors. In the Enlightenment, the scientific method and rational thought, creating a paradigm that separated scientific thinking (or philosophy) from religion. In Enlightenment discourse and ideas, science became central. Many Enlightenment writers and thinkers laid out principles in the sciences and associated scientific advancement with the virtues of religion and (rational authority) in favor of free speech and thought. Enlightenment writers emphasized rationalism, rejecting an idea of progress and achievement (James Turner). However, writers did not always agree to replace religion in the state with the sciences, questioning science among a non-Enlightened population and leaving the government for the development of the scientific profession. Enlightenment is fully not in conformity, its own, and authors marked the century as a period of political progress (Richard Schwab). The scientific discovery in nature, the development of biological theory, and new insights into chemistry and mechanics allowed for groundbreaking technological advances. The work of scientists like Galileo, Robert Hooke, and Thomas Copernicus will be covered in the article.

The Revolution's Effect in Astronomy

The major influence during the scientific and astronomical revolution, Nicholas Copernicus' work in the theories of astronomy, mathematics, and medicine, heavily influenced Copernicus' theory (The Scientific Revolution). The astronomical perspectives of the Late Middle Ages had been established by combining the theories of Aristotle, Ptolemy, and Claudius Ptolemy. As greater astronomical accuracy was demanded in the second century A.D. The universe was viewed as a collection of scientific systems with a hierarchy of fixed stars at their core. In the early Ptolemaic or geocentric conception, the world took up all the known elements of the universe, and earth, was considered the center and everything. The spheres that orbited the earth were composed of a transparent, crystalline material and moved in circles. As such, natural philosophers saw the new "perfect" type of science, unlike the "partial" natural bodies, which was thought to consist of an unchanging, immovable "perfectness" (Richard Schwab).

In 1710, Halley noted that Venus exhibited a complete range of phases, just as for other planets, which came up from Earth. This could be explained by the V system's orbit, which indicated in what phase of Venus would be observable during its orbital circumference around the sun, in contrast to the Ptolemaic model, which stipulated that only a few phases of Venus would be visible. Halley's observations of Venus and other planets made in Flamen's system, proving that these planetary mechanisms he built is different [Astronomical models](#).

Halley's Galileo was the first European scientist to use a telescope in making systematic observations of the heavens. Halley discovered variations in the speed, direction, and the phase of Venus.



Galileo Galilei and Physics Isaac Newton:

Isaac Newton was the first to try to derive mathematical predictions of natural phenomena.

How physical and physical causes, even though the nature of celestial bodies had been qualitatively presented in terms of physics and formed by mathematical theories of the Heliophysics and a Heliocentric in the Christian Harvener. As a result, the three laws of planetary motion that have the laws were discovered.

Isaac Newton's law of universal gravitation proved additional predictions Heliophysics and Heliocentric. Newton was able to explain all known gravitational motion in a single mathematical framework after realizing that the same laws that drive objects in the Earth's surface also kept the planets in orbit around it. For the first time, however, scientific understanding of the physical universe was shaped by Newton's Physics (1687), which established the principles of motion and universal gravity.

Newton developed an increasing skepticism regarding the viability of the Heliocentric model of the universe by substituting Copernicus' laws of planetary motion from his mathematical explanation of gravity and then applying the same ideas to explain planetary trajectories, tidal, equinox, precession, and other phenomena. This work also showed that the same ideas might be used to describe the motion of heavenly bodies and things on Earth. The law of universal gravitation, which merged terrestrial and celestial mechanics into a single, powerful system that appeared to be able to mathematically represent the entire world, was at some in the first time the most known. However, still the system was excluded, a relatively new branch of mathematics. This new science was created by Newton and his European contemporaries, described William Leake (1644-1716).

We must keep in mind that despite the great place in the pan of contemporary science, Isaac Newton was also deeply immersed in Greek philosophy, as shown by the hundreds of pages of papers he left behind dealing for example, Isaac Newton's masterpiece, *The Mathematical Principles of Natural Philosophy*, was published in 1687. What this work, Newton provided the new ideas that had been started by Copernicus, Kepler, and Galileo.

Additionally, he provided mathematical evidence in favor of his universal law of gravity.

Newton's research proved that the universe was a single, unified, homogeneous structure that's defined natural laws. He also explained the laws he used to arrive at his universal law.

Isaac Newton, Rules of Reasoning in Philosophy:

Rule 1:

The same cause of natural things should be admitted (first and best) that suffices to explain their phenomena.

Rule 2:

scared and angry, and the sickly black bile. Counting a patient's pulse became the primary diagnostic method since illness was thought to be caused by a harmful imbalance that could be identified from the distinct qualities of pulse. Pulse, which pulsates frequently, softened, heavy, weak, heaving, and flapping is critical for evidence. Individual human constitutions underwent increasingly diverse patterns.

Three physicians – Paracelsus, Wilhelm Vissler, and Wilhelm Harvey – are linked to the developments in medicine throughout the 16th and seventeenth centuries. Harman was born at a time in which Philippe Accordeau was. Harmanus (1593–1594) was the son of a wealthy dealer who experimented with astrology. He later changed his name to Paracelsus, after a more powerful than Galen, the ancient physician. Paracelsus smoothed much after leaving home in the age of Erasmus, and it is possible that he received medical degree from the University of Ferrara. In 1617, he had a total amount of time when he was named Harman's city physician and medical professor. His theories of his specific constitution, and that from this position, like so many others, didn't last.

Paracelsus developed a theory as basis of their theory, philosophy and suggested in both Aristotle and Galen's writings. His application and he helped to make the old system for a new chemical philosophy based on a direct interpretation of nature that came from new experiments and observations.

In fact, the chemical philosophy was actually related to a continuous process of college-based sciences that Paracelsus and his adherents were different from others who had employed chemical ideas in that they paid a less attention to the interpretation amount of their chemical's nature and more to that have been proposed. Because Paracelsus had abandoned the Galenic idea that "contraction" is a form of the old elements. With idea that "the more life," distorting the light amount was particularly critical. It takes in the light from the sun, the power that caused the illness would also be its remedy. Despite the burning efficacy Paracelsus had a self-question in the early century physicians. To some Paracelsus is primarily linked to his work in the diagnosis and treatment of plague. However, the outbreak actually was starting from the north of Vienna. Between 1614–1616, he and his writings on Galen, the medieval Islamic scholar, while studying medicine in Paris. Visser placed a strong emphasis on empirical development in the primary issues of understanding human anatomy, largely due to Galen's works associated with the Aristotelian Pythagoras.

He obtained a job as a professor of surgery in the University of Padua in 1596, having graduated with a PhD in medicine. His manuscript, *On the Nature of the Human Body*, was published in 1613. Visser was able to correct some of Galen's most obvious mistakes with his "hands-on" approach to anatomy. He insisted, he was unable to refute Galen's claim that the large blood veins come from the liver because his own studies clearly showed that they started from the heart. Visser's contribution continued to allow to correct of Galen's false claims, such as the Greek physician's theories regarding the ribs and flow of two types of blood in the veins and arteries. This theory reconstruction wasn't verified until William Harvey's research on blood circulation.

William Harvey (1578–1657) attended Cambridge University before continuing his education in Padua, where he graduated with a medical doctorate in 1602. After being sent to St. Bartholomew's Hospital in 1606, he worked as the physician to King James I and King Charles I. His first major work came with *The Motion of the Heart and Blood*, published in 1628. During the seventeenth century, Galen's physiological concepts were questioned, but his system remained largely unchanged.

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Harrison's experiments, which were based on seeing and observations, showed that as much death as the amount food is eaten. However, demonstrated that the heart, not the liver, is where the body's blood circulation begins. And the veins blood flows in body is not just arteries, veins – almost all – but the blood circulates in both in a group. Through the heart, Darwin's study seriously challenged Aristotle's view, but his theories about lungs, smoke, heat and the circulation, which explained how blood flows from the body's arteries to veins, were generally acknowledged in the 1800s.

Modern Science: Breakthroughs and Challenges

The modern field has been fundamentally shaped by the contemporary world with fast-changing technology, scientific literacy, and accumulated knowledge. Modern science evolution has been defined by advances across in the fields of medicine, artificial intelligence, space exploration, and quantum mechanics. With these developments, there has been a corresponding set of issues of ethical concerns, environmental, and accountability that define the future of scientific development.

One of the most significant scientific achievements of the past few decades is in medicine. The past work in heart and SA, vaccine technology, and medical research in the case of COVID-19 vaccines, demonstrates how the science of the future can benefit people global health issues in that new forms. Genetic engineering technologies like CRISPR is still there, some good progress in that they can fix health issues, holding out hope for potential cures for genetic-related diseases. Personalized medicine, treating individuals according to their unique genetic profile is another major finding that could answer revolution, enhancing treatment effectiveness and patient outcomes.

Artificial intelligence also has joined the ranks of scientific advancement. AI-based research results in drug discovery, advanced medical imaging diagnosis, and optimizing various physical and chemical processes. Machine learning algorithms are used in climate change models, forecasting, sustainable energy system optimization, and industry automation. AI also poses challenges in terms of data privacy, ethical usage, and algorithmic bias.

Space exploration has experienced an exponential transformation over the past few decades. With continued companies like SpaceX and Blue Origin at the forefront of creating reusable rockets, space travel is on the verge of becoming more feasible and accessible. The James Webb Space Telescope has provided unprecedented insights into the origins of the universe and has improved our understanding of cosmic phenomena. Furthermore, the efforts to harness Moon and extract minerals from lunar resources is just scientific dreams beyond the plane.

Quantum mechanics is also a very emerging area. Quantum computing is able to potentially revolutionize simulation processing by being as a platform to solve some types of problems much faster than conventional computers. The main challenge the high-energy community is the being very small. Several major goals and scaling up quantum technologies are still very formidable challenges to overcome and achieve.

Despite all these promising achievements, scientific research is confronted with a wide range of issues. Ethical concerns, particularly today's digital world, included scientific data manipulation, e.g. the ability of machines and global learning. AI will come in genetic research, artificial intelligence studies, and combination of computer and quantum data picked movement. Furthermore, scientific gaps in funding as well as access to education about the new science program universities will be bridging them.

In short, modern history of science has been characterized by groundbreaking findings that have revolutionized medicine, artificial intelligence, space exploration, and quantum mechanics. Future science, innovation, and

responsibility, however, is the responsibility assigned to the managers. With success coming forward, the credit and responsibility (success) will be assigned to persons in charge with doing the business.

Conclusion

In traditional and significant turning point in history, Western capitalism was the herald of the Revolution. The Protestant Reformation, rendered traditional into questions for the Western world during the Scientific Revolution, and a new understanding of the universe was developed, with the sense of order. The phrase "material belief" referring to a physical world, and an infinite rather than finite world. Changes in the idea of "truth" associated with challenge the idea of "human" European view, turned to reveal what they were as a part of the humanistic Revolution. The religious from medieval and a new started among people. Human on Earth need to be at the mercy of the universe. The earth was now seen as a small planet orbiting a sun that was simply a star in an infinite universe. Through the scientific revolution, human changed, the universe of people transformed their systems. But human not allowed that the universe also a big machine subject to natural laws, after all? The universal law of gravitation is the rule that the apple is attracted. When asked "how not threatened by reform?" We're not all human-made were perceived by